

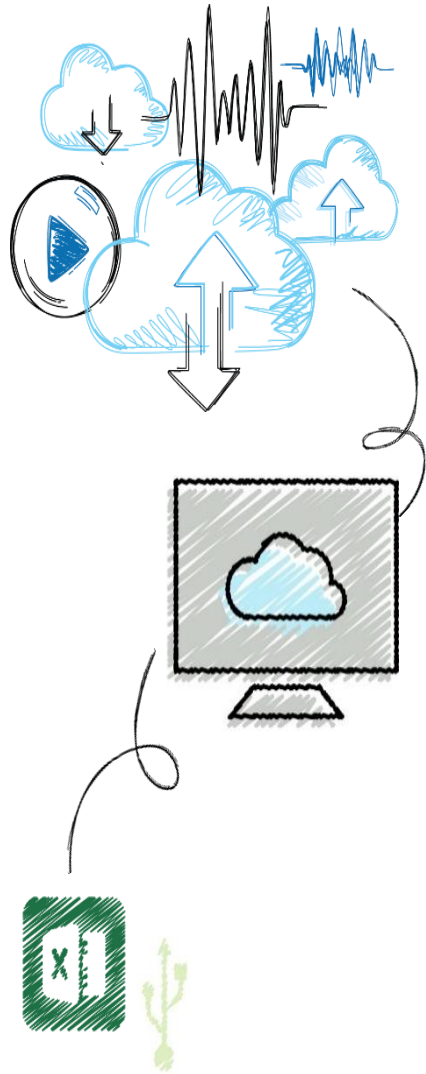
ICT703 Programming

Module 2 – Topic 1 Software Development



Objectives

1. Describe the basic phases of software development: analysis, design, coding, and testing
2. Review the 2 main methodologies for project development – Waterfall and Agile





The Software Development Process

- **Software development:** process of planning and organizing a program
 - Several approaches; one is the **waterfall model**
 - Waterfall model phases:
 - Customer request
 - Analysis
 - Design
 - Implementation
 - Integration
 - Maintenance
 - Modern software development is usually **incremental** and **iterative - Agile**
 - Analysis and design may produce a **prototype** of a system for coding, and then back up to earlier phases to fill in more details after some testing
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The Waterfall Method

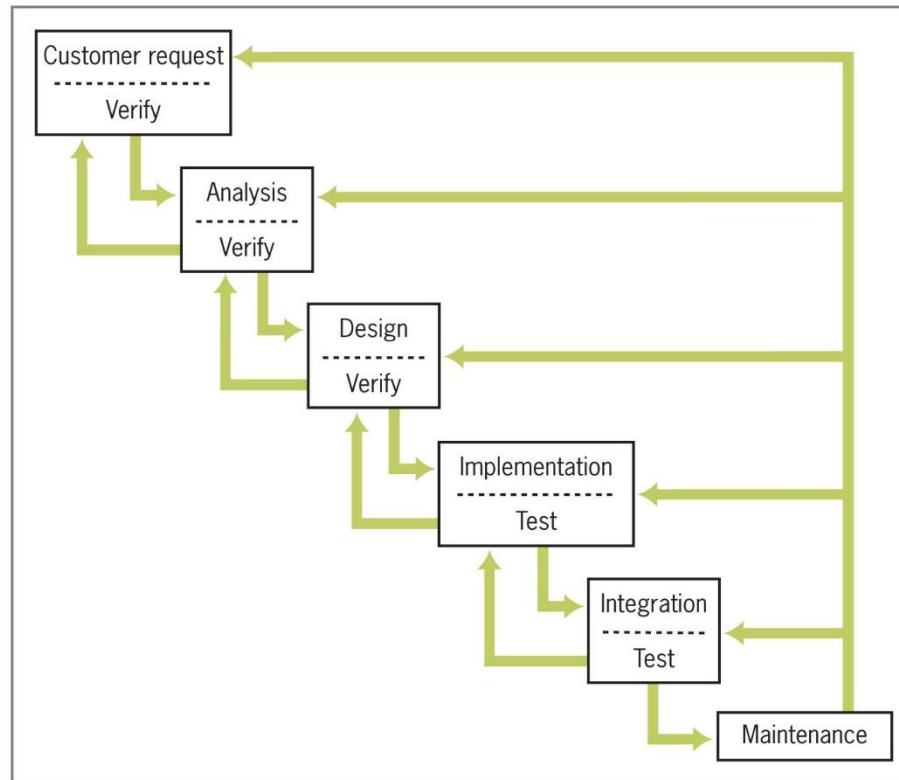


Figure 2-1 The waterfall model of the software development process



The Waterfall Method

- Programs rarely work as hoped the first time they are run
 - Must perform extensive and careful testing
 - The cost of developing software is not spread equally over the phases

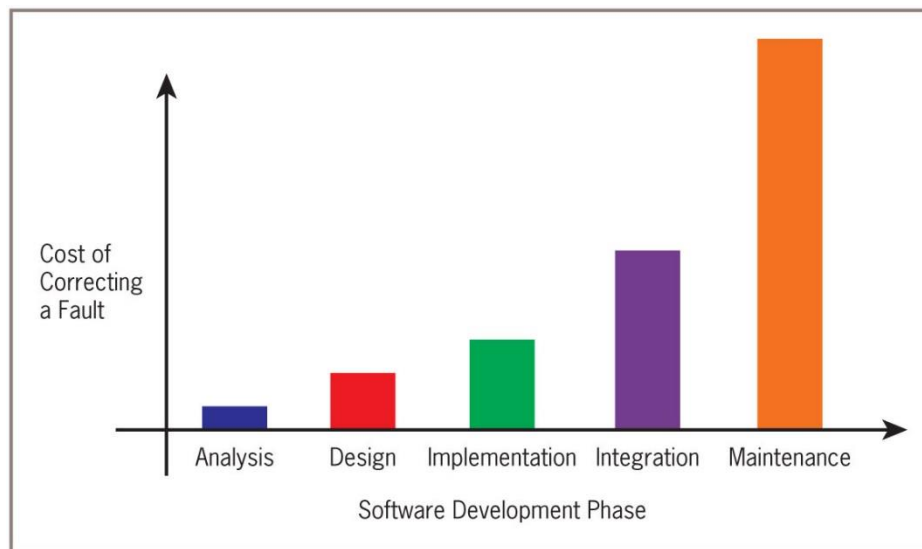


Figure 2-2 Relative costs of repairing mistakes that are found in different phases



The Waterfall Method

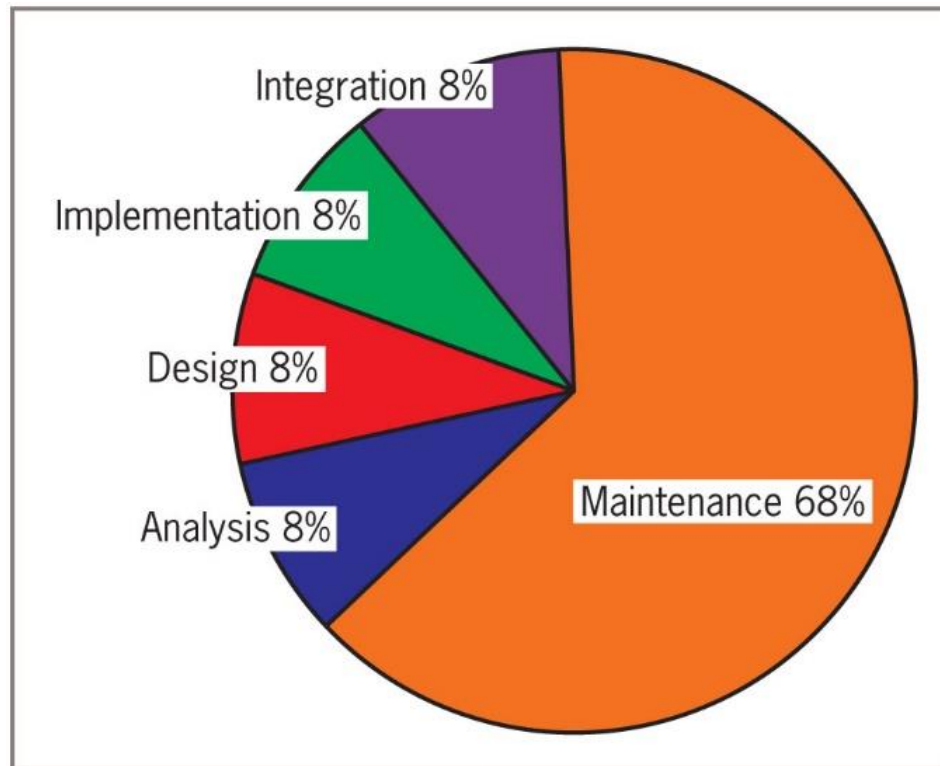
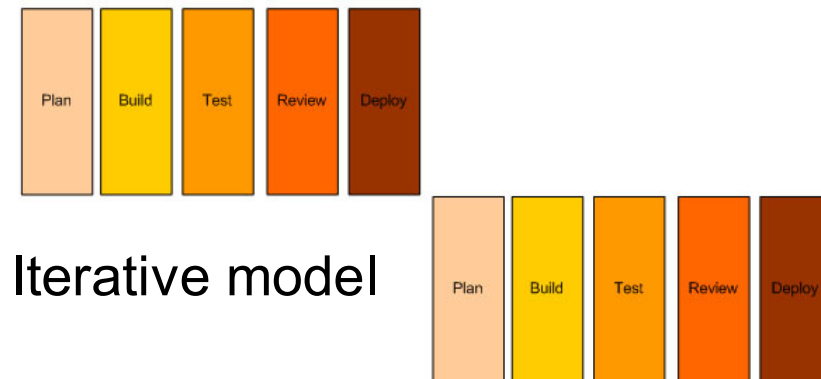
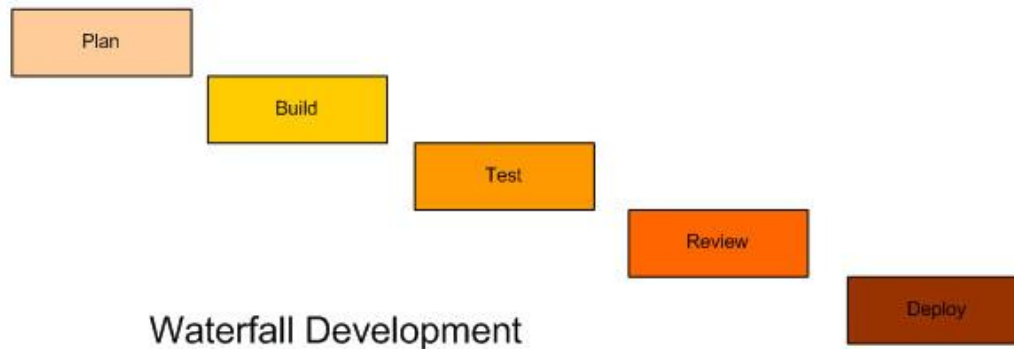


Figure 2-3 Percentage of total cost incurred in each phase of the development process



The Agile Method

- Based on the idea of delivering smaller functionality faster.





12 Agile Principles

- Satisfy the customer through early and continuous delivery
 - Welcome changing requirements, even late in development
 - Deliver working software frequently
 - Business people and developers must work together daily
 - Build projects around motivated individuals
 - At regular intervals, the team reflects
 - The most efficient and effective method is face-to-face conversation
 - Working software is the primary measure of progress
 - Agile processes promote sustainable development.
 - Continuous attention to technical excellence
 - Simplicity--the art of maximizing the amount of work not done--is essential.
 - The best designs emerge from self-organizing teams.
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One summary of differences

	Traditional	Agile
Fundamental Assumptions	Systems are fully specifiable, predictable, and can be built through meticulous and extensive planning.	High-quality, adaptive software can be developed by small teams using the principles of continuous design improvement and testing based on rapid feedback and change.
Control	Process centric	People centric
Management Style	Command-and-control	Leadership-and-collaboration
Knowledge Management	Explicit	Tacit
Role Assignment	Individual—favors specialization	Self-organizing teams—encourages role interchangeability
Communication	Formal	Informal
Customer's Role	Important	Critical
Project Cycle	Guided by tasks or activities	Guided by product features
Development Model	Life cycle model (Waterfall, Spiral, or some variation)	The evolutionary-delivery model
Desired Organizational Form/Structure	Mechanistic (bureaucratic with high formalization)	Organic (flexible and participative encouraging cooperative social action)
Technology	No restriction	Favors object-oriented technology